ENGR 852 Advanced Digital Design — Motion Estimator 14nm ASIC (520 MHz)
Complete Manual Reproduction Guide (Baby-Steps SOP)

This Standard Operating Procedure (SOP) documents every step required to reproduce the 520 MHz (1.923 ns clock period) SAED14nm Motion Estimator ASIC from scratch on a new MacBook Air connected to the SFSU Synopsys EDA server (mahmoodi.rcc.at.sfsu.edu). All architectural definitions align with Chapter 10, Section 10.4 (Example of Explicit Style—Motion Estimator, pp. 143–150) of the ENGR 852 course lecture notes.

1. Mathematical & Architectural Formulation (Chapter 10, Section 10.4)

• **Distortion Metric D(i,j)**: For a 16x16 reference macroblock R (r[m,n]) and a 31x31 search window S (s[m+i,n+j]), the Sum of Absolute Differences (SAD) for candidate index (i,j) in [0,15]x[0,15] is D(i,j) = Sum_{m=0..15} Sum_{n=0..15} |r[m,n] - s[m+i,n+j]|.
• **Coordinate Transformation**: Candidate indices (i,j) in [0,15]x[0,15] map to signed motion vector displacements (i-8, j-8) in [-8,+7]x[-8,+7] (8'hF8 to 8'h07 in 8-bit two's complement).
• **8-Bit Saturating PE Adder (Section 10.4.1, p. 147)**: Each PE computes {Carry, AccumulateIn} = Accumulate + difference; if (Carry == 1) AccumulateIn = 8'hFF. The Comparator initializes BestDist <= 8'hFF and updates BestDist, motionX, motionY only when newDist < BestDist (newBest = 1).
• **16-Bit Non-Saturating Datapath Extension**: Because 256 pixel differences of up to 255 sum to 256 x 255 = 65,280 (16'hFF00), widening Accumulate[15:0], PEout[255:0], and BestDist[15:0] (initialized to 16'hFFFF) eliminates intermediate 8-bit saturation across all 256 candidate blocks.

2. MacBook Air Terminal, VPN, and Passwordless SSH Setup

```
# 1. Open macOS Terminal (Cmd+Space -> Terminal) and install Xcode CLI tools:
xcode-select --install
mkdir -p ~/Developer/engr852-motion-estimator && cd ~/Developer/engr852-motion-estimator

# 2. Connect Palo Alto GlobalProtect VPN to portal: vpn.sfsu.edu

# 3. Generate Ed25519 SSH key and authorize on mahmoodi.rcc.at.sfsu.edu:
ssh-keygen -t ed25519 -C "engr852-macbook" -f ~/.ssh/id_ed25519 -N ""
cat ~/.ssh/id_ed25519.pub | ssh <SFSU_ID>@mahmoodi.rcc.at.sfsu.edu \
  "mkdir -p ~/.ssh && chmod 700 ~/.ssh && cat >> ~/.ssh/authorized_keys && chmod 600 ~/.ssh/authorized_keys"
ssh -Y <SFSU_ID>@mahmoodi.rcc.at.sfsu.edu
```

3. Modifying the Instructor's Original RTL and Testbenches for 520 MHz

3.1 Testbench Retargeting to 520 MHz (1.923 ns period): In top_tb.v and top_tb_direct.v, update timescale to 1ns/100fs, set half-period to #0.9615 (1.923 ns period = 520 MHz), use #0.02 post-posedge stimulus hold margin, and enable FSDB waveform dumping:

```
`timescale 1ns/100fs
always #0.9615 clock = ~clock; // 520.02 MHz (1.9230 ns period)
// Inside the 4112-cycle stimulus loop:
@(posedge clock);#0.02; R = Rmem[AddressR]; S1 = Smem[AddressS1]; S2 = Smem[AddressS2];
// Inside initial block for Verdi nWave:
$fsdbDumpfile("waves.fsdb");
$fsdbDumpvars(0);
```

3.2 16-Bit Non-Saturating Datapath & Cycle-4111 Terminal Hold (rtl/16bit/top.v): Extend PE Accumulate to [15:0], PEout to [255:0], Comparator BestDist to [15:0] (initialized to 16'hFFFF), and hold Control count at 13'h100F (4111 cycles = 16*256 + 15 Rpipe fill cycles, Table 10.1, p. 146):

```
// In module PE (rtl/16bit/top.v):
AccumulateIn = Accumulate + {8'd0, difference};
if (newDist == 1) AccumulateIn = {8'd0, difference};

// In module Control (terminal count hold at 13'h100F = cycle 4111):
if (count == 13'h100F) completed <= 1'b1;
else count <= count + 13'd1;

// In module Comparator:
```

```
if (CompStart == 0) BestDist <= 16'hFFFF;
else if (newBest == 1) begin BestDist <= newDist; motionX <= ...; motionY <= ...; end
```

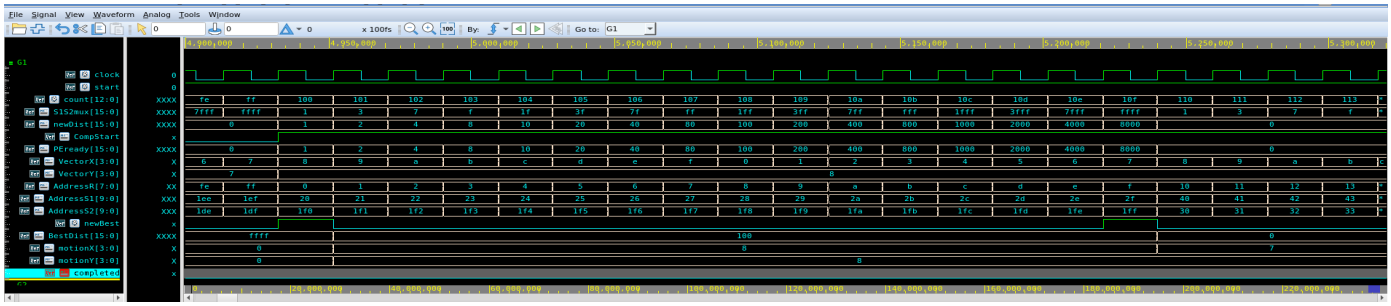
Testbench	Stimulus	8-Bit Result	16-Bit Result	Technical Mechanism (Ch. 10, Sec. 10.4)
top_tb_direct.v	ref.txt & search.txt exact match at (-8,-8)	BestDist = 8'h00 X=F8, Y=F8 (-8,-8)	BestDist = 16'h0000 X=F8, Y=F8 (-8,-8)	Zero pixel difference across all 256 pixels: D(0,0) = 256 x 0 = 0.
top_tb.v	Random Rmem + 8'h01 perturbation at (-5,+7)	BestDist = 8'hFF X=FA, Y=00 (-6,0)	BestDist = 16'h0100 X=FB, Y=07 (-5,+7)	D(3,15) = 256 x 1 = 256 (16'h0100). 8-bit PE saturates at 8'hFF and ties strict '<'.
top_tb_16bit.v	Rmem=FF, Smem=00 at (-5,+7); 1-LSB at (-8,-8)	Saturates at 8'hFF on all 256 blocks	BestDist = 16'hFF00 X=FB, Y=07 (-5,+7)	Verifies upper bound: 256 x 255 = 65,280 (16'hFF00) < 16'hFF01 selects (-5,+7).

4. Synopsys VCS Simulation & Verdi nWave Waveform Inspection

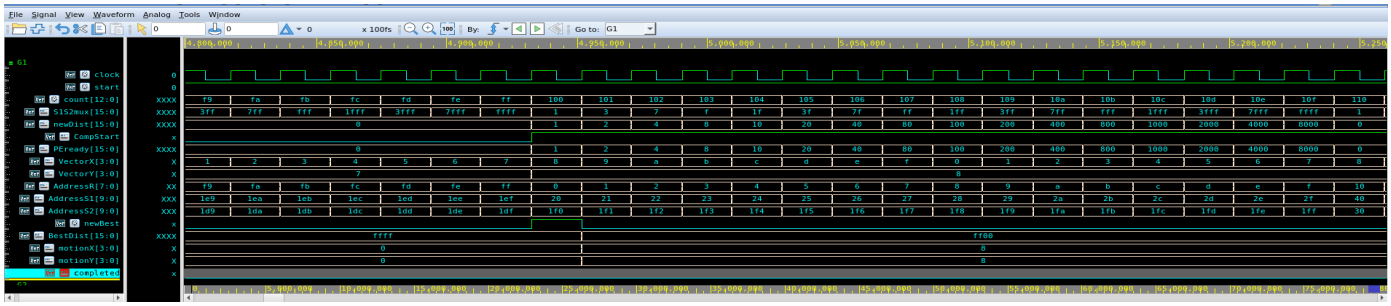
```
source /apps/settings
vcs -full64 -debug_access+all rtl/16bit/top.v tb/16bit/top_tb_direct.v -o simv_direct && ./simv_direct
vcs -full64 -debug_access+all rtl/16bit/top.v tb/16bit/top_tb.v -o simv_random && ./simv_random
vcs -full64 -debug_access+all rtl/16bit/top.v tb/16bit/top_tb_16bit.v -o simv_diag && ./simv_diag
verdi -ssf waves.fsdb &
```

How to Capture Both Full-Overview and Zoomed-In Hex Value Waveforms in Verdi nWave:
1. Press **g** (Signal -> Get Signals), select clock, start, BestDist, motionX, motionY, AddressR, AddressS1, AddressS2, R, S1, S2, and u1/u2/count, and click **Add**.
2. Press **f** (Zoom Full) to verify the complete 4,111-cycle execution window (0 ns to 7,905.45 ns) with the full horizontal scroll bar.
3. Drag a zoom box across 7,895 ns to 7,910 ns (or press **z**) so Verdi renders the individual hex values (count = 100f, BestDist = 0000 / 0100 / ff00, motionX = fb, motionY = 07) inside each bus bar.

Captured Verdi nWave Full + Zoomed-In Hex Waveform (Direct Testbench D(15,0) = 16'h0000):



Captured Verdi nWave Full + Zoomed-In Hex Waveform (Maximum Distortion D(0,0) = 16'hFF00):



5. Synopsys Fusion Compiler (14nm P&R), IC Validator DRC, and 3-Corner PrimeTime STA

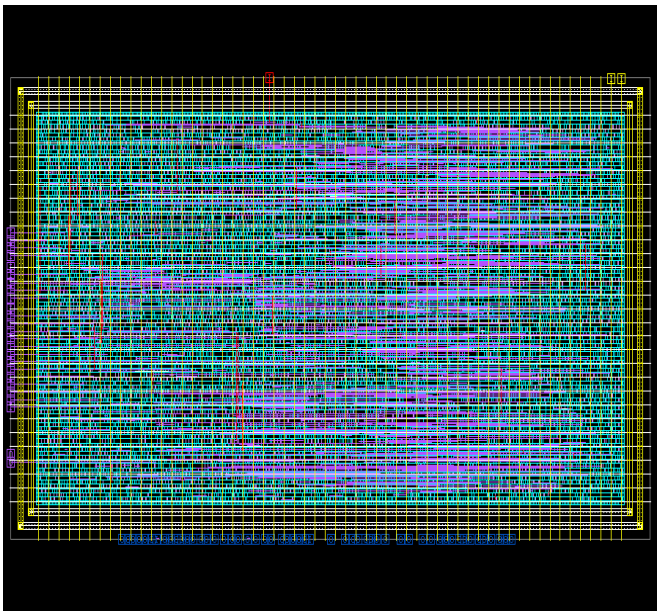
```
# 1. Run SAED14nm Synthesis, Floorplan (125.07 um x 62.80 um), Placement, CTS, Route, SPEF & GDS:
fc_shell -f scripts/server/run_fc_16bit_520mhz.tcl | tee logs/fc_16bit_520mhz.log

# 2. Run Sign-Off Design Rule Check (IC Validator -> 0 / 242 rule violations):
icv -vue -i results/top.gds -c top -f GDSII \
    /apps/synopsys/SAED14_EDK/tech/icv_drc/saed14nm_1p9m_drc_rules.rs

# 3. Run Standalone 3-Corner PrimeTime STA (FAST, TYPICAL, SLOW setup & hold verification):
pt_shell -f scripts/server/run_pt_3corner_520mhz.tcl | tee logs/pt_3corner_520mhz.log
```

Metric	Final Sign-Off Value (V9 14nm 520 MHz)	Verification Status
Clock Period / Frequency	1.923 ns (520.02 MHz, 0.20 ns uncertainty, 0.50 ns I/O delay)	VERIFIED (top_520mhz.sdc)
Placed Cell Count / Area	8,230 logic cells (449 FFs) / 3,416.18 um^2 (0 SRAM/ROM macros)	VERIFIED (Fusion Compiler)
Die / Core Dimensions	125.07 x 62.80 um (7,854.40 um^2 die; 6,558.99 um^2 core)	VERIFIED (52.08% utilization)
PT Setup WNS (FAST / TYP / SLOW)	+0.71 ns / +0.54 ns / +0.22 ns (0 violations)	VERIFIED (3-Corner SPEF STA)
PT Hold WNS (FAST / TYP / SLOW)	+0.00 ns / +0.04 ns / +0.14 ns (0 violations)	VERIFIED (3-Corner SPEF STA)
IC Validator DRC & LVS	0 / 242 DRC rule violations; 0 LVS errors; 0.890 mW TT power	VERIFIED (DRC & LVS CLEAN)

Captured SAED14nm Routed Layout (top.gds, 242/242 ICV DRC Clean):



6. Pushing the Complete Project to a New GitHub Repository

```
cd ~/Developer/engr852-motion-estimator
git init -b main
git add .
git commit -m "Complete 520 MHz 14nm Motion Estimator ASIC RTL, Physical Design, and Sign-Off Package"
git remote add origin https://github.com/<GITHUB_USERNAME>/motion-estimator-asic-lab-notebook-2026.git
git push -u origin main
```